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Cenozoic northward translation of the Kitakami massif in northeast Japan: paleomagnetic evidence

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Abstract

Paleogene and early Cretaceous welded tuffs have been sampled from eleven sites at the Omoe Peninsula (39.6°N, 142.0°E) in the Kitakami massif of northeast Japan for paleomagnetic study. Characteristic paleomagnetic directions with a high unblocking temperature component above 560°C are isolated: Westerly declinations with shallow inclination are identified after tilt correction in the 62–71 Ma Heizaki volcanics ($D = 283.5°$, $I = 9.4°$, $\alpha_{95} = 8.0°$) and in the 114–119 Ma Harachiyama Formation ($D = 274.2°$, $I = -20.4°$). The shallow inclination, which was previously reported from early Cretaceous plutonic rocks without tilt correction in the Kitakami massif, has been discovered in the tilt-corrected paleomagnetic directions. The presence of both normal and reversed polarities suggests reliability of this shallow and westerly paleomagnetic direction. The shallow inclination indicates that the Kitakami massif was located at low latitude ($5° \pm 4°$N) at 71–62 Ma. Taking into account a reference paleomagnetic pole of the Paleocene expected from Sikhote Alin, the Kitakami massif was translated northward over 25° in latitude since 65 Ma. The Kitakami massif can be subjected to latitudinal northward motion more than 24° during the last 70 Ma, when it was transported by push of the moving Pacific plate and reached the Asian continental margin near Sikhote Alin later than 30 Ma. We conclude that the Kitakami massif was incorporated into northeast Japan at some time between 30 and 22 Ma after the Cenozoic northward translation. © 1997 Elsevier Science B.V.

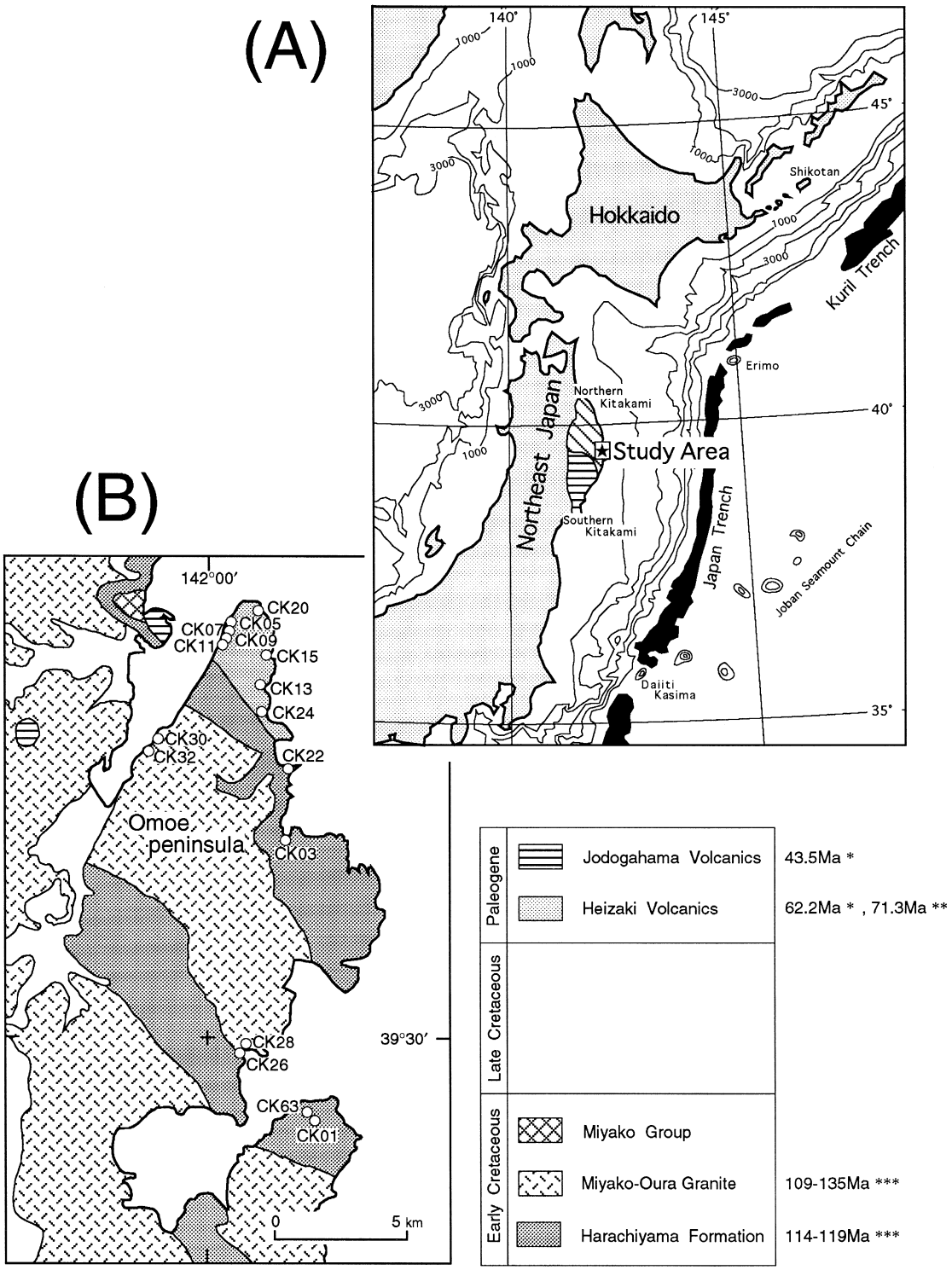
Keywords: paleomagnetism; tectonics; Cenozoic; Japan; Pacific Plate; accretion

1. Introduction

The Cenozoic northward motion of continental fragments has been recognized in the northeastern part of the Pacific Ocean for more than two decades. Tectonic collage of terranes is observed from southeast Alaska to Oregon along the northeastern part of the Pacific Ocean. The assemblage of terranes comprises continental fragments with different geological histories [1,2]. Paleomagnetism and age determination revealed that these fragments were situated at lower latitude in Mesozoic times (e.g., [3]). These fragments had been transported on the Pacific plate from a southerly paleo-position and reached nearly in place with respect to cratonic North America in Cenozoic time. Cenozoic translation of terranes and their accretion to the craton are not rare events in the northeastern part of the Pacific Ocean.

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(A)
140°
145°
45°
40°
35°
1000
3000
Hokkaido
Shikotan
Kuril Trench
Erimo
Northern Kitakami
Study Area
Northeast Japan
Southern Kitakami
Japan Trench
Joban Seamount Chain
Daiiti Kasima
(B)
142°00′
39°30′
CK20
CK07
CK05
CK11
CK09
CK15
CK13
CK24
CK30
CK32
CK22
Omoe peninsula
CK03
CK28
CK26
CK63
CK01
0
5 km
Paleogene
Late Cretaceous
Early Cretaceous
Jodogahama Volcanics
43.5Ma *
Heizaki Volcanics
62.2Ma * , 71.3Ma **
Miyako Group
Miyako-Oura Granite
109-135Ma ***
Harachiyama Formation
114-119Ma ***

Few Cenozoic northward motions of terranes have been reported in the northwestern part of the Pacific Ocean. The island arc system from the Kuril to the northeast Japan arcs forms a paired morphology of an outer non-volcanic arc and an inner volcanic arc. The Habomai Island–eastern Nemuro is the outer arc of the southern part of the Kuril arc, and the Kitakami mountain area is an outer non-volcanic province in the northeast Japan arc (Fig. 1). Since the outer arcs are located in front of the consumed zone of plates, these are plausible blocks which experienced Cenozoic translation and consequently accreted to the continent. Recently, Bazhenov and Burtman [4] postulated Miocene accretion of Shikotan to the Inner arc of the Kuril arc after its northwestward motion.

We have undertaken the paleomagnetic study of the Kitakami massif in the Japan arc to investigate the possibility of its Cenozoic northward motion, because the Kitakami massif has been recognized to be an allochthonous block [5]. Previous paleomagnetic study of the Kitakami massif revealed that some early Cretaceous granitic rocks distributed in the northern part of the Kitakami massif preserved shallow inclination ($I \approx 20°$) in remanent magnetization [6]. Because no tectonic correction is applied to these granitic rocks, this anomalously shallow inclination is attributed to local block tilting. The welded tuffs of the Paleogene Heizaki volcanics and the Cretaceous Harachiyama Formation are our present target, because bedding planes were well preserved as eutaxitic structure (lineation of stretched pumice and aligned phenocrysts).

2. Geological setting and sampling

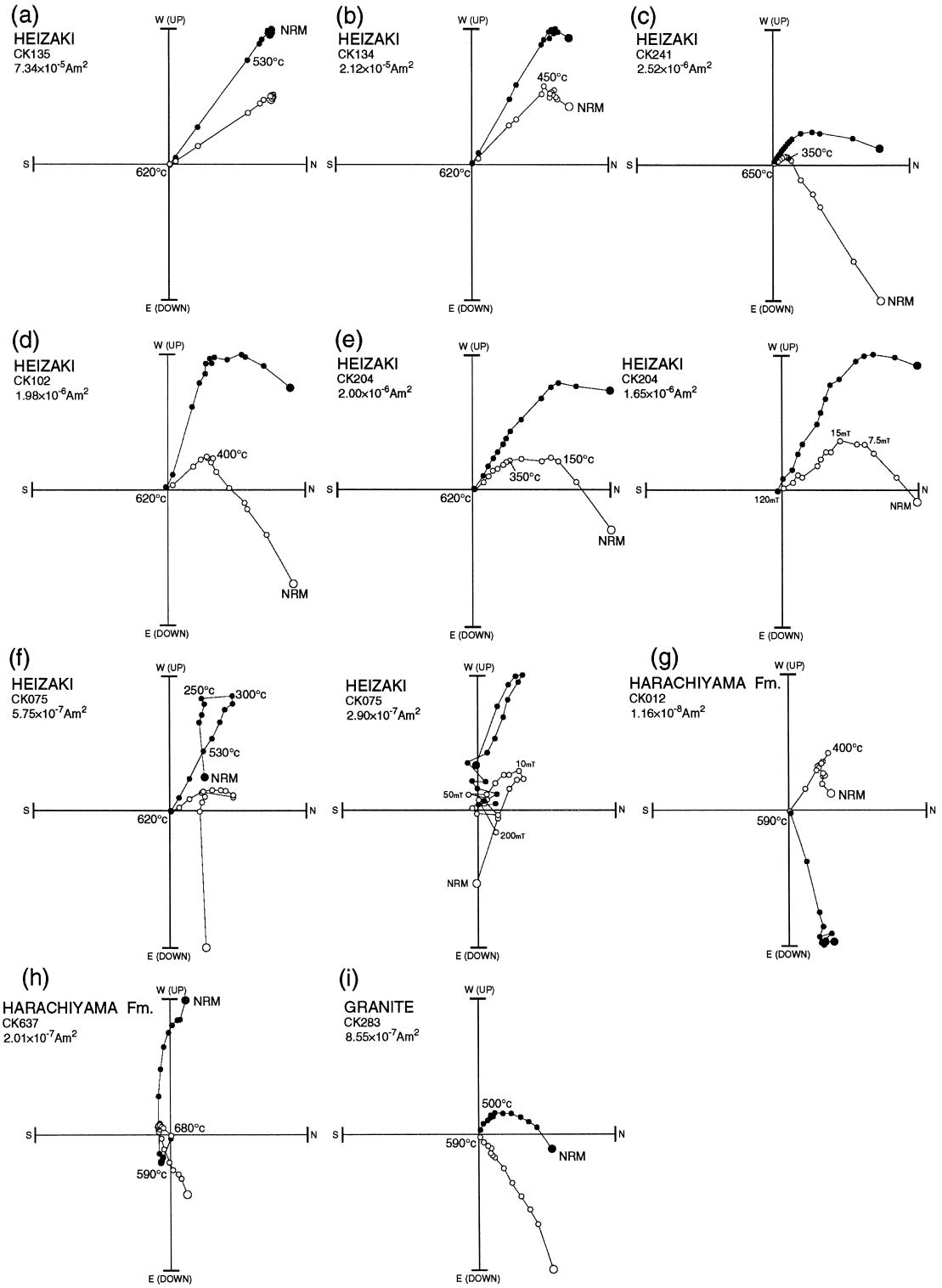
Two geological provinces are arranged in northern Honshu. Mesozoic and older rocks form the Kitakami massif which is situated in the Pacific Ocean side (outer arc) (Fig. 1), whereas the remaining area is characterized by Neogene volcano-sedimentary rocks (inner arc).

Pre-Tertiary rocks are widely exposed in the Kitakami massif [7,8]. The Northern Kitakami terrane comprises the Jurassic accretionary complexes and the Southern Kitakami terrane is characterized by a pre-Cretaceous shallow marine sequence. Two terranes are connected by the Hayachine tectonic belt consisting of mafic to ultramafic rocks. Late Mesozoic magmatism began in the early Cretaceous throughout the Southern and Northern Kitakami terranes: Hauterivian to Barremian andesite–rhyolite volcanisms and granitic magmatism (110–125 Ma) occurred.

Volcanic rocks of Cretaceous and Paleocene times are distributed around the Miyako district which is situated on the southeastern part of the Northern Kitakami terrane [9,10] (Fig. 1B). Middle early Cretaceous magmatism formed the Harachiyama Formation (114–119 Ma) as well as the Miyako-Oura granite (109–135 Ma) [11]. The Harachiyama Formation is intruded by the Miyako-Oura granite and overlain by the Miyako Group. The Paleogene magmatism occurred in Paleocene and Eocene times, and produced the Jodogashima acidic volcanics and Heizaki volcanics. The latter is exposed at the northern edge of the Omoe Peninsula. It extruded onto the late-Aptian to Albian Miyako Group and has yielded radiometric ages of 62.2 ± 2.5 Ma (K–Ar date) [12] and 71.3 ± 2.4 Ma (^{40}Ar–^{39}Ar date) [13]. These volcanics are composed of dacitic volcanocrastic rocks associated with welded tuffs. The succession dips gently from 13° to 48° eastward.

The welded tuffs in the Harachiyama Formation and the Heizaki volcanics rocks were collected at eleven sites as hand samples oriented by magnetic compass. Granitic rocks were also collected at five sites in the Omoe Peninsula. Each paleomagnetic site typically comprised of twelve independently oriented samples, distributed over distances ranging up to 15 m. Since a eutaxitic structure was identified at each

Fig. 1. (A) Location map of study area and a tectonic unit of the Kitakami massif consisting of the northern and southern terranes. (B) Geological map of study area showing stratigraphic relationship in the Omoe Peninsula of the Northern Kitakami terrane. Map is adapted from Geological Map of Iwate Prefecture [9] and Geology of the Miyako district [10]. The sampling site locations are marked by *open circles*. Ages are after (*) Uchiumi et al. [12], (**) Takigami [13] and (***) Shibata et al. [11].


(a)
HEIZAKI
CK135
7.34×10-5Am2
W (UP)
NRM
530°c
S
N
620°c
E (DOWN)
(b)
HEIZAKI
CK134
2.12×10-5Am2
W (UP)
450°c
NRM
S
N
620°c
E (DOWN)
(c)
HEIZAKI
CK241
2.52×10-6Am2
W (UP)
350°c
S
N
650°c
NRM
E (DOWN)
(d)
HEIZAKI
CK102
1.98×10-6Am2
W (UP)
400°c
S
N
620°c
NRM
E (DOWN)
(e)
HEIZAKI
CK204
2.00×10-6Am2
W (UP)
150°c
350°c
S
N
620°c
NRM
E (DOWN)
HEIZAKI
CK204
1.65×10-6Am2
W (UP)
15mT
7.5mT
S
N
120mT
NRM
E (DOWN)
(f)
HEIZAKI
CK075
5.75×10-7Am2
W (UP)
250°c
300°c
530°c
NRM
S
N
620°c
E (DOWN)
HEIZAKI
CK075
2.90×10-7Am2
W (UP)
10mT
50mT
S
N
200mT
NRM
E (DOWN)
(g)
HARACHIYAMA Fm.
CK012
1.16×10-8Am2
W (UP)
400°c
NRM
S
N
590°c
E (DOWN)
(h)
HARACHIYAMA Fm.
CK637
2.01×10-7Am2
W (UP)
NRM
S
N
680°c
590°c
E (DOWN)
(i)
GRANITE
CK283
8.55×10-7Am2
W (UP)
500°c
S
N
590°c
NRM
E (DOWN)

outcrop of the welded tuffs, post-emplacement tilting of the welded tuffs was adjusted for on the basis of this.

3. Paleomagnetism

In the laboratory, individual specimens, 25 mm in diameter and 23 mm long, were prepared from these samples. Natural remanent magnetizations (NRMs) were measured with either a spinner magnetometer (Natsuhara SMM-85) or a cryogenic magnetometer (ScT and 2G) according to the intensity of the magnetization. Specimens were thermally demagnetized using a laboratory-built furnace: the residual magnetic field in which the specimens were cooled is < 10 nT. AF demagnetization was carried out with a three-axis tumbler system in three cylindrical mu-metal shield. The thermal demagnetization method was more effective than the AF demagnetization for erasing the secondary magnetization (Fig. 2), so that all the remaining specimens were subjected to stepwise thermal demagnetization up to 680°C. Results for each specimen were plotted on orthogonal vector diagrams [14] to assess component structure as well as on equal-area projections to evaluate directional stability. Principal component analysis [15] was used to estimate the component directions. Progressive acquisition of isothermal remanent magnetization (IRM) and thermal demagnetization of a three-component IRM [16] were performed to identify ferromagnetic mineral content (Fig. 3).

3.1. Welded tuffs in the Heizaki volcanic rocks

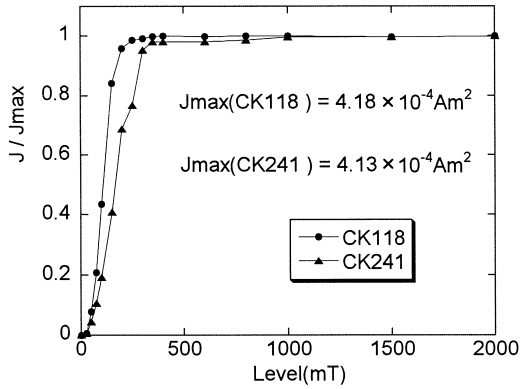
Demagnetization behavior strongly depends on NRM intensity of a sample. Samples with NRM intensity of 7 to 1 A/m (sites of CK11, CK13 and CK15) often revealed a single well defined component of magnetization during progressive thermal demagnetization (Fig. 2a). A large percentage of magnetization is lost by 580°C but a small remanent magnetization with unblocking temperatures $>$ 580°C remains. Both magnetite and hematite probably carry the stable remanent magnetization with almost the same direction, which is confirmed by thermal demagnetization of the three-component IRM (Fig. 3A).

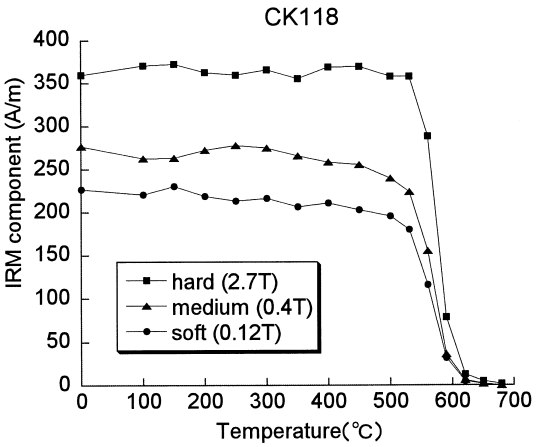
Samples with NRM intensity between 5 and 0.2 A/m from 5 sites (sites of CK09, CK11, CK13, CK15 and CK24) had generally two-component magnetization (Fig. 2b and c), whereas samples with NRM intensity of < 0.2 A/m revealed three to four components (CK05, CK07 and CK20) (Fig. 2d–f). The highest-temperature component appears after demagnetization at 300–450°C and reveals straightforward demagnetization behavior with unblocking temperature between 590° and 630°C. Thermal demagnetization of the three-component IRM suggests presence of magnetic mineral fraction with unblocking temperature at 200–300°C and at 580–620°C (Fig. 3A; CK241 and CK204). The former fraction is possibly responsible for the low-temperature component. The highest-temperature component resides probably in titanium-poor magnetite and hematite. Direction of the high-temperature component is recognized as a characteristic one for each sample.

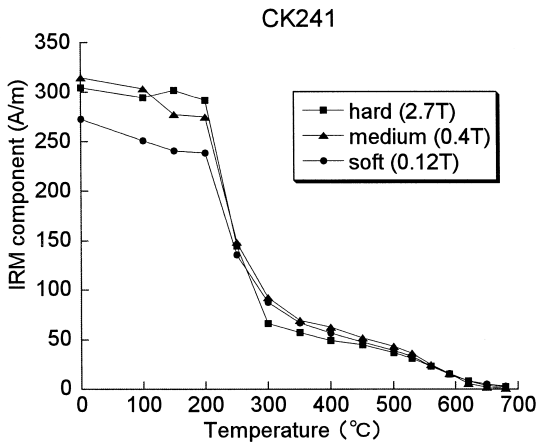
The high-temperature component directions are clustered tightly in all the seven sites except for CK20: precision parameter ranges from 17.4 to 235.5 (Fig. 4a; Table 1). Site mean paleomagnetic directions fall around a shallow and large westerly direction. The mean direction of the seven sites is $D = 285°$, $I = -17°$ in situ, and $D = 284°$, $I = 9°$ after tilt correction. No fold test is possible since the bedding attitude shows homoclinal. Although the precision parameter decreases from 131 to 58 after tilt correction, the decrease is insignificant for a negative ‘‘fold test of McElhinny [17]’’ even at the 95% level. This increase in dispersion is possibly due to uncertainty in strike measurements (of the

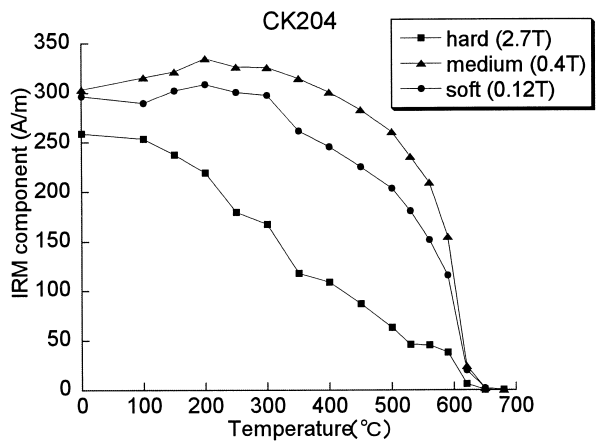
Fig. 2. Orthogonal projections of magnetization vector end-points during thermal and AF demagnetization experiments for specimens: Heizaki volcanics of sites CK13 (a, b), CK24 (c), CK10 (d), CK20 (e) and CK07 (f); Harachiyama Formation of sites CK01 (g) and CK63 (h); and Miyako-Oura granite of site CK28 (i). *Solid symbols* for horizontal projection, *open symbols* for vertical. Tectonic tilt correction has not been applied.

(A)

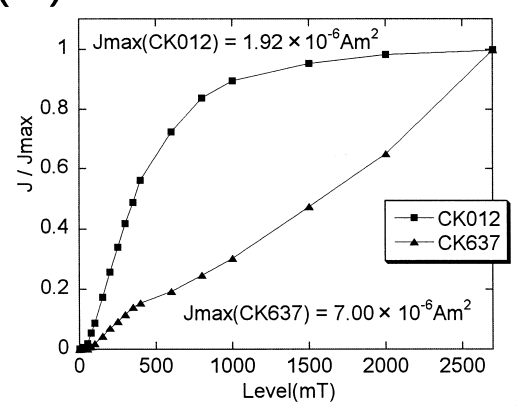

J / Jmax
Jmax(CK118) = 4.18 × 10-4Am2
Jmax(CK241) = 4.13 × 10-4Am2
CK118
CK241
Level(mT)


CK118
IRM component (A/m)
hard (2.7T)
medium (0.4T)
soft (0.12T)
Temperature(℃)


CK241
IRM component (A/m)
hard (2.7T)
medium (0.4T)
soft (0.12T)
Temperature (℃)


CK204
IRM component (A/m)
hard (2.7T)
medium (0.4T)
soft (0.12T)
Temperature(℃)

(B)


J / Jmax
Jmax(CK012) = 1.92 × 10-6Am2
Jmax(CK637) = 7.00 × 10-6Am2
CK012
CK637
Level(mT)

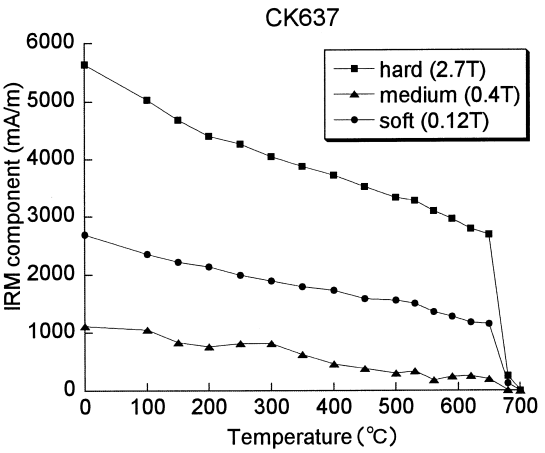

CK637
IRM component (mA/m)
hard (2.7T)
medium (0.4T)
soft (0.12T)
Temperature (℃)

Table 1
Paleomagnetic results for the Paleogene–early Cretaceous igneous rocks from the Omoe Peninsula in northeast Japan

Sitename	*N*	*n*	Intensity (A/m)	Demagn.(°)	In-situ		Tilt-corrected		*k*	α_{95}(°)	Lat.(°N)	Long.(°E)
					D (°)	*I* (°)	*D* (°)	*I* (°)				
Heizaki volcanic rocks (62.2 Ma, 71.3 Ma [12,13]):												
CK05	12	10	~ 0.1	300–620	281.5	−9.6	281.0	7.3	105.4	4.7	39°38.6′	142°00.6′
CK07	12	9	~ 0.05	300–620	282.4	−12.0	281.9	−0.9	34.0	9.0	39°38.5′	142°00.6′
CK09	12	11	~ 0.5	350–620	280.0	−10.3	282.0	18.7	217.9	3.1	39°38.4′	142°00.5′
CK11	12	10	~ 5	300–620	284.2	−22.1	278.1	−5.9	235.5	3.2	39°38.3′	142°00.5′
CK13	12	12	~ 3	300–620	290.9	−22.3	288.7	10.2	239.1	2.8	39°37.4′	142°01.5′
CK15	12	12	~ 1.5	300–620	285.8	−21.1	285.7	20.6	17.4	10.7	39°38.2′	142°01.8′
CK20 *	12	12	~ 0.2	450–650	300.7	−3.5	305.1	30.6	17.1	10.8	39°38.9′	142°01.5′
CK24	11	11	~ 0.2	450–650	288.7	−21.5	287.1	15.9	30.9	8.4	39°36.8′	142°01.6′
Mean:												
(In-situ)	7				284.7	−17.0			131.1	5.3	39°38′	142°0.1′
(Tilt-corrected)	7						283.5	9.4	58.0	8.0	39°38′	142°0.1′

VGP												
					Lat.	Long.	Lat.	Long.		*A*95		
(In-situ)	7				45.8°E	5.6°N				3.7°		
(Tilt-corrected)	7						47.4°E	13.5°N		4.7°		
Harachiyama Formation (114–119 Ma [11]):												
High-temperature component												
CK01	11	10	~ 0.001	350–620	79.9	6.3	89.2	12.1	15.4	12.7	39°29.0′	142°02.5′
CK63	11	9	~ 0.001	530–680	101.3	−15.6	99.7	28.9	45.7	7.7	39°29.0′	142°02.5′
Mean	2				90.4	−4.7	94.2	20.4			39°29.0′	142°02.5′
Medium-temperature component												
CK63M	11	10	~ 0.001	100–560	276.3	13.3	273.7	−29.6	53.7	6.7	39°29.0′	142°02.5′
CK03M	11	8	~ 0.001	100–500	285.9	21.2	282.9	1.1	52.7	7.7	39°34.3′	142°02.3′
Mean	2				281.0	17.3	283.4	−14.3			39°29.0′	142°02.5′
Miyako-Oura Granite (109–135 Ma [11]):												
CK26	12	7	~ 0.1	450–620	282.7	41.7			27.2	11.8	39°31.2′	142°02.4′
CK28	12	11	~ 0.1	450–620	317.6	49.3			15.7	11.9	39°31.1′	142°02.3′

Magnetization directions of the high- and medium-temperature component are listed. *N* and *n* are numbers of field oriented samples and samples used for statistics, respectively. Intensity is approximate value of remanent magnetization before demagnetization. Demagn. is temperature range used to calculate the characteristic direction by principal component analysis in °C. *D* and *I* are declination and inclination, respectively; *k* is the Fisherian precision parameter; α^{95} is the radius of cone of 95% confidence within site.

Fig. 3. Progressive acquisition of IRM and thermal demagnetization of a three-component IRM produced by applying a different DC field (2.7, 0.4 and 0.12 T) to each of the three perpendicular axes of the sample.
(A) Heizaki volcanics. All samples reach saturation IRM in 0.3 T in DC field. Their three-component IRM demagnetization shows that the unblocking temperature of all coercivity components ranges between 580° and 630°C, indicating that all samples are dominated by magnetite and hematite. Thermal demagnetization in all coercivity components of CK241 (site CK24) and in the hard component of CK204 (CK20) suggests presence of magnetic mineral fraction with unblocking temperature around 300°C.
(B) Harachiyama Formation. Sample CK637 from site CK63 does not reach saturation IRM in 2.7 T in DC field. IRM and its thermal demagnetization shows that sample CK637 is dominated by hematite.

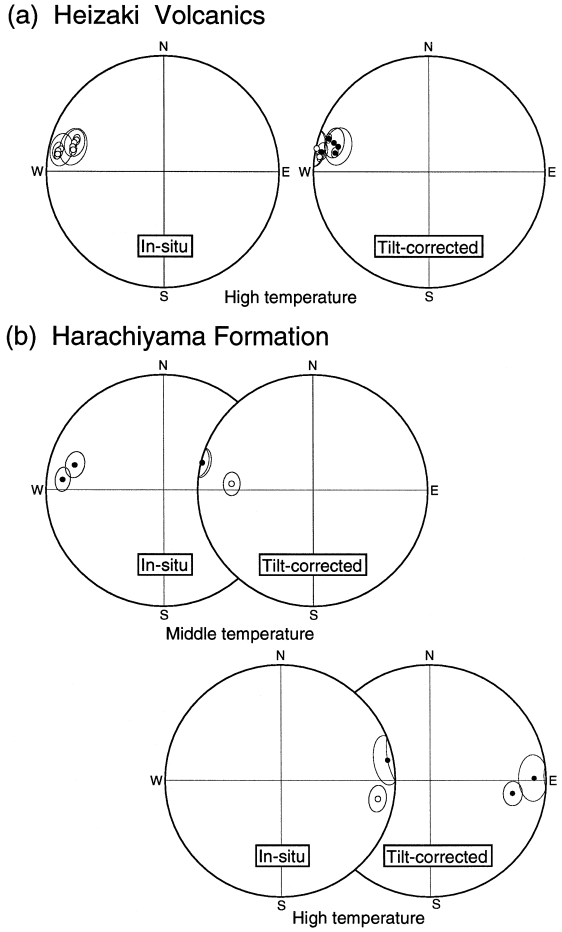


Fig. 4. Equal-area projections of site mean directions with a circle of 95% confidence.
(a) High-temperature component of the Heizaki volcanics before and after tilt correction.
(b) High- and medium-temperature component of the Harachiyama Formation, before and after tilt correction.
The *solid symbols* refer to the lower hemisphere, the *open symbols* refer to the upper hemisphere.

order of $\pm 5°$) for estimating of paleohorizontal plane from the eutaxitic structure in the welded tuffs.

Site CK20 preserves remagnetization phenomena (Fig. 5). The high-temperature components are dispersed along a great circle (Fig. 5c): Tilt-corrected high-temperature component directions are distributed from a shallow and westerly direction ($D = 281°$, $I = 4°$) to a steep and northwesterly direction ($D = 332°$, $I = 52°$). The high-temperature component is possibly a resultant magnetization between the characteristic paleomagnetic vector of the Heizaki volcanics ($D = 284°$, $I = 9°$) and the medium-temperature component of this site (in situ; $D = 310°$, $I = 24°$) (Fig. 5b). This site is, therefore, excluded for calculating a mean paleomagnetic direction of the Heizaki volcanic rocks.

3.2. Welded tuffs in the Harachiyama Formation

Samples had fairly weak NRM intensity between 3×10^{-3} and 0.3×10^{-3} A/m. Samples of three sites had variable components within one site — single, two and three components (Fig. 2g and h). IRM and its thermal demagnetization show that the samples of CK01 and CK03 are dominated by mag-

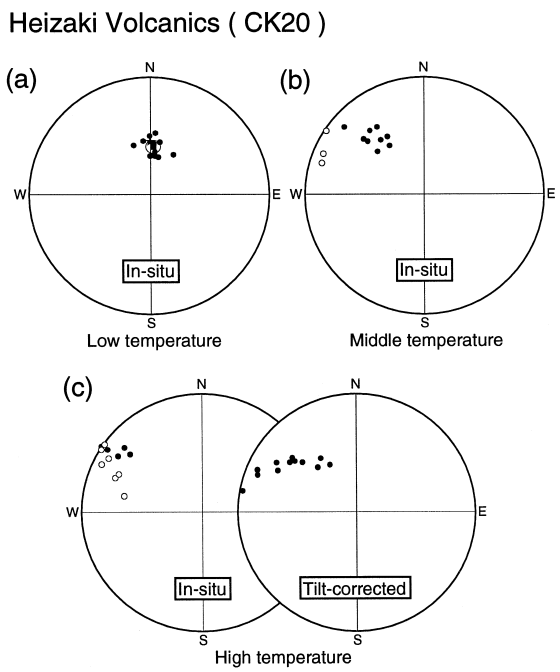


Fig. 5. Equal-area projections of paleomagnetic directions for three temperature components of specimens from CK20 in the Heizaki volcanics.
(a) Low-temperature component (in situ) with a mean paleomagnetic direction (■) and a circle of 95% confidence. ★: present dipole field direction. The low-temperature component directions in situ are distributed around the present dipole field ($D = 3.3°$, $I = 56.6°$).
(b) Medium-temperature component (in situ). The medium-temperature component directions of 8 in 12 specimens cluster around a northwesterly direction with moderately steep inclination in situ ($D = 320.6°$, $I = 38.1°$).
(c) High-temperature component before and after tilt correction.
The *solid symbols* refer to the lower hemisphere, the *open symbols* refer to the upper hemisphere.

netite and hematite, and the samples of CK63 by hematite (Fig. 3B).

Straightforward demagnetization behavior to the origin is identified for two sites of CK01 and CK63: unblocking temperature is $\sim 620°$ and $\sim 680°C$, respectively. The highest-temperature component magnetization reveals large clockwise deflection in declination with shallow inclination ($D = 94.2°$, $I = 20.4°$) after tilt correction (Fig. 4b).

Medium-temperature component is extracted from two sites of CK03 and CK63. Linear segments of three or more points is observed on an orthogonal vector plot for a temperature range between 100° and 560°C. This paleomagnetic direction shows counter-clockwise deflection in declination ($D = 281.0°$, $I = 17.3°$) in situ (Fig. 4b).

3.3. Granites

Samples with fairly strong NRM intensity (~ 0.1 A/m) revealed straightforward behavior after thermal demagnetization above 500°C (Fig. 2i). This high-temperature component with unblocking temperature of $\sim 600°C$ is identified from two sites (CK26, CK28) and shows moderately steep inclination with northwesterly declination ($D = 290–320°$, $I \approx 45°$). Samples with weak NRM of < 0.05 A/m revealed only spurious remanent magnetization.

4. Discussion

4.1. Shallow inclination of the Paleogene paleomagnetic direction

Shallow inclination with large counter-clockwise deflection in declination characterizes the Paleogene Heizaki volcanics of 62–71 Ma. The shallow direction in remanent magnetization is obtained after high-temperature thermal demagnetization and after tilt correction. The tilt-corrected mean direction is $D = 283.5°$, $I = 9.4°$ and $\alpha^{95} = 8.0°$ (Fig. 6a). No

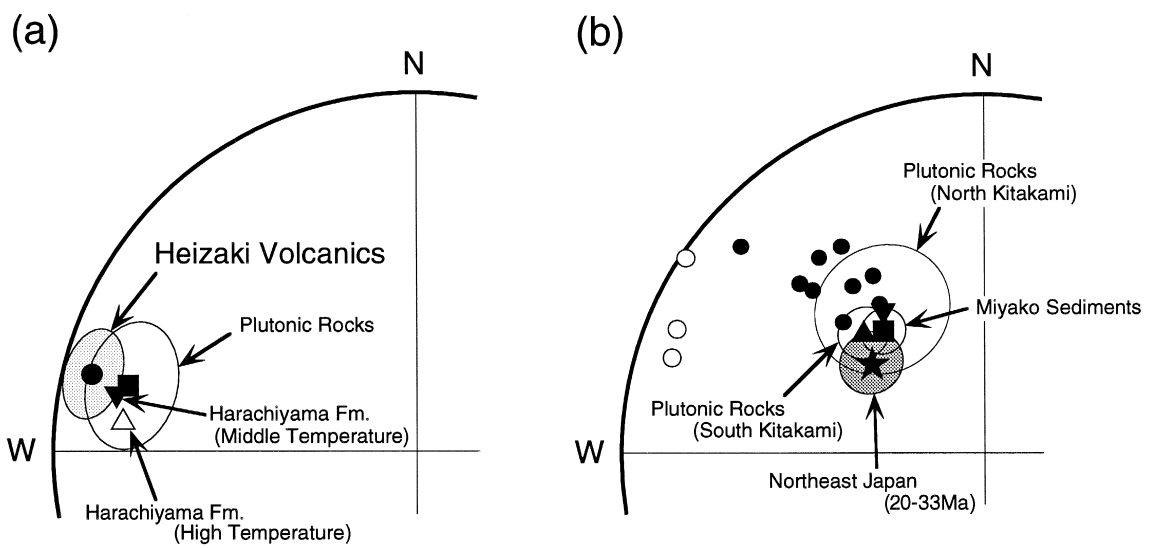


Fig. 6. Equal-area projections of paleomagnetic directions (a) with shallow inclination for the Northern Kitakami terrane and (b) with steep inclination for northeast Japan.
(a) ●: Heizaki volcanics (the high-temperature component after tilt corrrection) with circles of 95% confidence (shown in *shaded pattern*). The Harachiyama Formation [▲: high-temperature component after tilt correction (the direction is inverted through the origin), ▼: medium-temperature component before tilt correction]. ■: plutonic rocks (before tilt correction) with circles of 95% confidence [6,18].
(b) ★: The tilt-corrected 20–33 Ma paleomagnetic directions of northeast Japan with circles of 95% confidence (shown in *shaded pattern*) [25]. ■: Miyako Cretaceous sediment (in situ) with circles of 95% confidence [18], ▲ and ▼: plutonic rocks (in situ) with circles of 95% confidence from Northern and Southern Kitakami terranes, respectively [6,18,22]. Medium-temperature component of CK20 is drawn by *circles*.
The *solid symbols* refer to the lower hemisphere, the *open symbols* refer to the upper hemisphere.

remagnetization phenomena are observed in the Heizaki volcanics except for CK20, and the magnetization component carried by the magnetic mineral fraction with unblocking temperature around 300°C is completely erased. The in-situ mean direction is far from the present-day dipole field. We believe that this component magnetization is of primary origin.

The shallow inclination is also observed in the Harachiyama volcanics (114–119 Ma) (CK01 and CK63) (Fig. 6a). The high-temperature component after tilt correction shows shallow inclination with large clockwise deflection in declination ($D = 94.2°$, $I = 20.4°$), being almost antipodal to the direction of the Heizaki volcanics. This direction is recognized as a reversed polarity of the paleomagnetic direction of the Heizaki volcanics. The shallow inclination with counter-clockwise deflection in declination is also observed in the medium-temperature component of CK03 and CK63; in situ ($D = 281.0°$, $I = 17.3°$) and after tilt correction ($D = 283.4°$, $I = -14.3°$).

The shallow and westerly paleomagnetic direction has also been reported from Cretaceous plutonic rocks distributed in the Northern Kitakami terrane [6] (Fig. 6a). Two sites of gabbros and six sites of granites show large westerly declination and shallow inclination. Both normal and reversed directions are recognized; a mean direction is $D = 283°$, $I = 20.5°$, $\alpha^{95} = 12°$ [18]. Although the ages of these plutonic rocks (100–121 Ma) [11,19] are older than that of the Heizaki volcanics and their paleomagnetic directions are obtained without tilt correction, the mean paleomagnetic direction of these plutonic rocks is in good agreement with that of the Paleocene Heizaki volcanics.

Four different types of remanent magnetizations record the shallow and westerly paleomagnetic direction accompanied with normal and reversed polarities: the high-temperature component of the Heizaki volcanics, the high- and medium-temperature components of the Harachiyama Formation, and the NRM of the Cretaceous plutons. This suggests that the shallow inclination is not spot data but is a characteristic paleomagnetic direction of fairly long time between the early Cretaceous and Paleocene in the Northern Kitakami terrane.

Moderately steep inclination with northwesterly declination ($D \approx 320°$, $I \approx 55°$) has been believed to be a characteristic paleomagnetic direction for pre-40 Ma rocks of the Kitakami massif since 1961 [20] (Fig. 6b). This direction was observed in in-situ paleomagnetic directions of Miyako Cretaceous sediments [18,21] and the early Cretaceous plutonic rocks in the Northern Kitakami terrane [22], and in in-situ paleomagnetic directions of the Cretaceous plutonic rocks [6,22] and the Devonian to Jurassic sedimentary rocks [21,23] in the Southern Kitakami terrane. The present study shows that a similar direction appears in the in-situ direction of the high-temperature component of the Miyako-Oura granite (CK28) and the medium-temperature component of CK20 of the Heizaki volcanics in the Northern Kitakami terrane. These in-situ Paleozoic to Paleogene paleomagnetic directions are in excellent agreement with the tilt-corrected 20–33 Ma paleomagnetic directions of northeast Japan ($D = 308.8°$, $I = 57.2°$, $\alpha^{95} = 7.2°$; [24]) (Fig. 6b). As Moreau et al. [18] postulated, this component of the Paleozoic to Paleogene rocks is ascribable to a secondary origin which was acquired in 30–20 Ma. We conclude that welded tuffs in the Omoe Peninsula preserve an older shallow paleomagnetic direction which precedes the northwesterly and steep paleomagnetic direction.

The shallow inclination of $< 15°$ contrasts well with moderate deep inclination values reported from Paleogene rocks in the eastern margin of the Asian continent. Recent data from the Paleogene welded tuffs in Sikhote Alin (51 Ma Bogopol Group and 66 Ma Sijanov Group) provide fairly steep inclination values of $\sim 58°$ [25]. Tertiary and upper Cretaceous poles for the North China block (86.9°N, 255.7°E [25] and 86.3°N, 200.1°E [26]) yield steep inclination of 58° for the Sikhote Alin area. The steep inclination is also reported from Paleogene welded tuffs of the southwest Japan arc: The welded tuffs with 60 Ma for the San'in area shows 52° in inclination [27] and the late Cretaceous Nohi rhyolite shows 50° [28]. The persistently shallow inclination of the Northern Kitakami terrane can hardly be explained by paleogeomagnetic behavior (e.g., excursion or reversal).

4.2. Tectonic implications of the shallow inclination

The shallow inclination of the Northern Kitakami terrane is attributed to northward translation of the whole Kitakami massif from a southern place. The

northward translation model of the Kitakami massif is concordant with its allochthonous nature in the Japan arc [5]. The shallow inclination of the Heizaki volcanics corresponds to paleolatitude of 5 ± 4°N at 62–71 Ma, and Harachiyama Formation yields paleolatitude of 6.1°S and 15.4°S for the period 114–119 Ma (Fig. 7A). Expected paleoposition for the Kitakami massif at 62–71 Ma is calculated on the basis of its paleolatitude and the 51–66 Ma paleomagnetic pole position from Sikhote Alin (Fig. 7B). The Kitakami massif was translated northward over 25° in latitude since 65 Ma (Fig. 7A).

The paleomagnetic data claim that the Kitakami massif approached the Asian continental margin near Sikhote Alin at some time between 30 and 22 Ma. Fairly steep inclination ($I = 42°$) with large westerly declination [24] is observed in the Nisatai Formation with K–Ar ages of 21.8 and 21.0 Ma [29,30] which is distributed on the northern part of the Northern Kitakami terrane. The Paleozoic to Paleogene rocks of the Kitakami massif secondarily acquired moderately steep inclination ($I = 54.5°$) with northwesterly declination in 30–20 Ma [18] (Fig. 7A). The large westerly declination of the Kitakami massif indicates that it has already reached the Asian continental margin by ~ 20 Ma and was subjected to counterclockwise rotation as a part of the northeast Japan arc associated with opening of the Japan Sea [31,32].

The amount and timing of the northward motion of the Kitakami massif is well explained by a model in which it is transported by plate motion of the Pacific plate. Trajectory of the northward movement of the Pacific plate was evaluated by backward modeling [33,34] on the basis of the plate tectonic model in the Pacific Ocean [35] (Fig. 7B). Assuming that the Kitakami massif was transported by push of the moving Pacific plate, this trajectory corresponds to a locus of the northward translation of the Kitakami massif. Latitudinal northward motion between 24° and 35° can be accomplished for the Kitakami massif during the last 70 Ma as it reached the Asian continental margin near Sikhote Alin at a period between 30 and 0 Ma. The 70 Ma paleoposition for the Kitakami massif on the trajectory of the 30 Ma arrival model is close to the 62–71 Ma paleoposition expected from paleomagnetic study (Fig. 7B). Agreement of the paleopositions improves when the Kitakami massif arrived at the Asian conti-

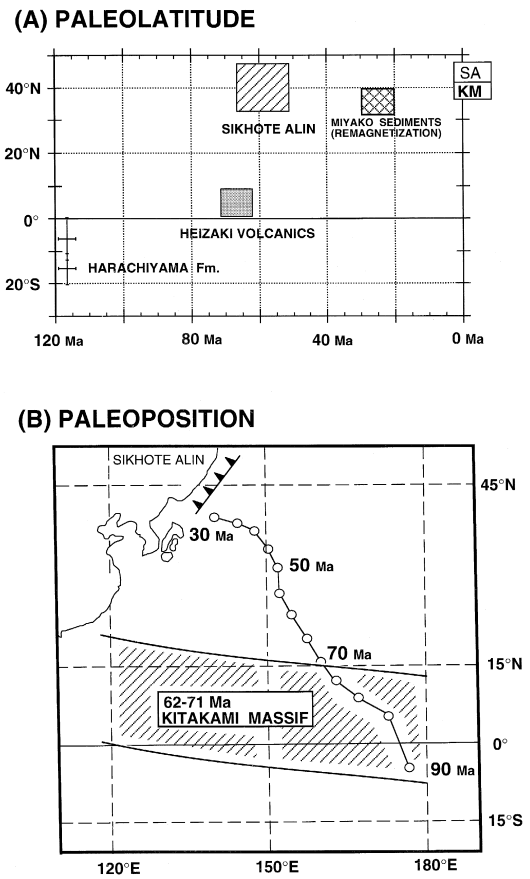


Fig. 7. Paleolatitude and paleoposition of the Kitakami massif.
(A) Paleolatitude with 95% confidence limit determined by the paleomagnetic study for the Heizaki volcanics and Harachiyama Formation of the Northern Kitakami terrane (present study), and for the Sikhote Alin [25]. Inclination of remagnetized Miyako Cretaceous sediments [18] provides paleolatitude at a period between 30 and 20 Ma for the Kitakami massif. Range in age for the Heizaki volcanics is from the highest and lowest radiometric ages. *KM* and *SA*: present latitude for the Kitakami massif and the Sikhote Alin, respectively.
(B) Paleopositions of the Kitakami massif are expected from both paleomagnetic viewpoints and the plate tectonic model. *Hatched region* represents locus of the paleomagnetically determined paleolatitude of the Kitakami massif at 62–71 Ma with 95% confidence limit relative to the paleomagnetic pole position of the Sikhote Alin at 51–66 Ma [25]. The confidence limit includes uncertainty in both the paleolatitude and the paleomagnetic pole. *Open circles* are a trajectory of the Kitakami massif moving with push by the Pacific plate and arriving in the Asian continental margin at 30 Ma [34]. Southwest Japan prior to opening of the Japan Sea is reconstructed after Otofuji and Matsuda [27]. An oceanic plate to the northwest of the Kitakami massif was engulfed in a subduction (*barbed line*) along the eastern Asian continental margin.

nental margin at a period between 30 and 22 Ma as predicted by paleomagnetism.

This model accounts for similarities in low inclination and westerly declination between the Kitakami massif and the Cretaceous to Paleogene seamounts on the Pacific plate along the Japan trench. The Cretaceous Katori Seamount and the Joban Seamount chain show fairly low inclinations ranging between 7.8° and −19.5° [36,37]. Inclination values of Erimo Seamount (104–79 Ma) and seamounts of Daiiti–Kasima (120–105 Ma) are estimated to be 22–36° [38–40] and 18.3° [36,41], respectively. The Kitakami massif as well as these seamounts was transported by the Pacific plate from its birth place of low latitude. As systematic counter-clockwise deflection in declination of the seamounts [36,37,39,40] originates in the motion of the Pacific plate around the Pacific–Kula Euler pole between 85 and 43 Ma [35,42]; this motion partly contributes to the large amount of westerly declination of the Northern Kaitakami terrane.

The regional geology appears to indicate the Cenozoic approach of the Kitakami massif to the Asian continent. First of all, a subduction-related accretionary prism (Taukha terrane) at the end of the early Cretaceous has been discovered in Sikhote Alin [43]. Because the Kitakami massif was located at lower latitude far away from Sikhote Alin at 119–114 Ma, an oceanic plate to the northwest of the Kitakami massif was engulfed in a subduction along the eastern Asian continental margin. The accretionary prism, therefore, formed at the edge of the Asian continent. Secondly, the Kitakami massif overrode an old subduction zone along Sikhote Alin when it juxtaposed to Sikhote Alin at 30–22 Ma. Consequently a new subduction formed at the present Japan trench. The subduction formation gave rise subsidence of a insular high (the Oyashio landmass) at the shelf edge off the northeast Japan arc since the end of Oligocene (25 Ma) [44,45]. Lastly, the subduction jump explains migration of the volcanic area in northeast Japan [46–49]. Extensive volcanic activity from 80 to 30 Ma occurred in the Sikhote Alin belt [50] and in Japan Sea side of the northeast Japan arc [48]. The volcanic activity area migrated from the Japan Sea side to the Pacific Ocean side at the early Miocene (∼22 Ma).

Northward motion and Miocene accretion of a land block is postulated for Shikotan island in the Lesser Kuril Islands [4]. This block overrode the subduction zone from the Pacific plate to the Eurasian continent in Miocene time. Neogene northward motion of blocks and their accretional phenomena occurred in the central part of Japan (the Izu-Bonin arc) [51,52] and in the eastern part of Taiwan (the Luzon arc) [53] associated with the northward migration of the Philippine Sea plate [54,55]. Cenozoic accretion events of blocks associated with the northward motion of the oceanic plate are possibly fairly common phenomena in the northwestern part of the Pacific Ocean as well as in northeastern part of the Pacific.

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